

NFS Setup Documentation (RHEL 9 Server + RHEL 8 Client)

Server Configuration (RHEL 9)

Set static IP (server side)

```
[root@nfssrv1 srv]# nmcli connection modify ens160 ipv4.method manual ipv4.addresses "192.168.0.100/24" ipv4.gateway "192.168.0.1"
```

```
[root@nfssrv1 srv]# nmcli connection up ens160
```

Set hostname

```
[root@nfssrv1 srv]# hostnamectl set-hostname nfssrv1.radical.com
```

```
[root@nfssrv1 srv]# hostname  
nfssrv1.radical.com
```

Create repo file

```
vi /etc/yum.repos.d/local.repo
```

```
[BaseOS]  
name=BaseOS  
baseurl=file:///run/media/root/RHEL-9-5-0-BaseOS-x86_64/BaseOS  
enabled=1  
gpgcheck=0  
  
[AppStream]  
name=AppStream  
baseurl=file:///run/media/root/RHEL-9-5-0-BaseOS-x86_64/AppStream  
enabled=1  
gpgcheck=0
```

Install NFS packages

```
[root@nfssrv1 srv]# yum install -y nfs-utils.x86_64
```

Enable NFS services

```
root@nfssrv1 srv]# systemctl enable --now nfs-server
```

```
[root@nfssrv1 srv]# systemctl enable --now rpcbind
```

```
[root@nfssrv1 srv]# systemctl status nfs-server
```

Create shared directory

```
[root@nfssrv1 srv]# mkdir -p /srv/nfs/share/
```

```
[root@nfssrv1 srv]# chmod 777 /srv/nfs/share/
```

Configure export

```
vi /etc/exports
```

```
File Edit View Search Terminal Help
/srv/nfs/share 192.168.0.100/24(rw,sync,no_root_squash)
```

Apply exports

```
[root@nfssrv1 srv]# exportfs -rav
exporting 192.168.0.100/24:/srv/nfs/share
```

```
[root@nfssrv1 srv]# exportfs -v
/srv/nfs/share 192.168.0.100/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,no_root_squash,no_all_squash)
```

Configure firewall

```
[root@nfssrv1 srv]# firewall-cmd --permanent --add-service=nfs
success
```

```
[root@nfssrv1 srv]# firewall-cmd --permanent --add-service=mountd
success
```

```
[root@nfssrv1 srv]# firewall-cmd --permanent --add-service=rpc-bind
success
```

```
[root@nfssrv1 srv]# firewall-cmd --reload  
success
```

```
[root@nfssrv1 srv]# firewall-cmd --list-services  
cockpit dhcpv6-client mountd nfs rpc-bind ssh
```

SELinux configuration

```
[root@nfssrv1 srv]# setsebool -P nfs_export_all_rw 1  
[root@nfssrv1 srv]# chcon -Rt public_content_rw_t /srv/nfs/share
```

Client Configuration (RHEL 8)

Install NFS client package

```
[root@localhost ~]# yum install -y nfs-utils
```

Create mount directory

```
mkdir -p /mnt/nfs
```

Test NFS exports

```
[root@localhost ~]# showmount -e 192.168.0.100  
Export list for 192.168.0.100:  
/srv/nfs/share 192.168.0.100/24
```

Manual mount test

```
[root@localhost ~]# mount 192.168.0.100:/srv/nfs/share /mnt/nfs
```

Verify mount

```
[root@localhost ~]# df -h
```

```
[root@localhost ~]# mount | grep nfs
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
192.168.0.100:/srv/nfs/share on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,
rsize=262144,wsize=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,
sec=sys,clientaddr=192.168.0.10,local_lock=none,addr=192.168.0.100)
```

Make mount permanent (fstab)

```
vi /etc/fstab
```

```
192.168.0.100:/srv/nfs/share /mnt/nfs nfs defaults 0 0
```

Reload systemd and mount all

```
systemctl daemon-reload
```

```
[root@localhost ~]# mount -a
```

Verification Test

Create file on server

```
[root@nfssrv1 srv]# echo "NFS KUNAL TEST FILE" > /srv/nfs/share/test.txt
```

Check on client

```
[root@localhost ~]# cat /mnt/nfs/test.txt
NFS KUNAL TEST FILE
```

Result Achieved

RHEL 9 server exports directory via NFS

RHEL 8 client mounts remote filesystem

Shared storage accessible across network

Persistent mount configured via `/etc/fstab`

Firewall + SELinux configured

